Keys.

CpE 3110/CS3803 Homework Assignment #5 Maximum – 100pts

You must always show or explain your work in a neat and orderly format. You are encouraged to discuss ideas with other students and consult references, but your work must be your own.

1. Suppose you are given two different LEGv8 processors, P1 and P2, namely. P1 is a single-cycle processor with clock cycle time of 800ps. P2 is a **7-stage** pipelined processor with the clock cycle time of 200ps. If you are given a program with 1000 instructions without any pipeline hazards, calculate the execution times on P1 and P2. Then, calculate the speedup (i.e., the ratio of execution times of P1 and P2) [25pts].

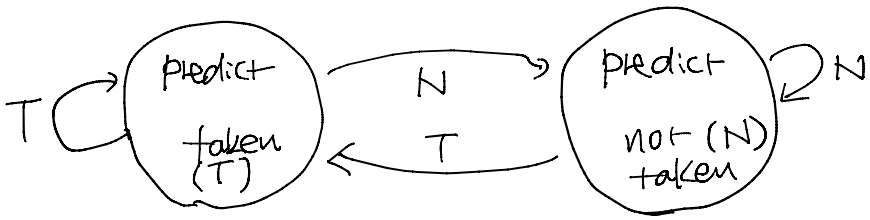
2. Suppose you are given a 5-stage pipelined LEGv8 processor with both EXE and MEM forwarding paths added. Consider the following instruction sequence [30pts]:

```
ADD X9, X9, X10
ADD X9, X9, X11
LDUR X9, [X9, #0]
ADD X9, X9, X12
```

a) Clearly identify all data hazards.

b) For each identified data hazard, which forwarding path should be used? Do you need to insert any stall clock cycle(s)? Justify your answer.

3. Suppose you are given a pipelined LEGv8 processor with built-in 1-bit dynamic predictor as shown below. Calculate the branch prediction accuracy (= # of correct predictions / # of total predictions) of the following branch outcome sequence. Assume that the initial state is "Predict taken" [30pts].



Branch outcome sequence (T means "branch-taken" and N means "branch-not-taken"):

T, T, N, N, N, T, N, T, N, T, T, N, N

4. For the branch outcome sequence given in question 3, calculate the branch prediction accuracy of the static branch prediction strategies including "Always predict taken" and "Always predict not taken". Among three branch prediction

strategies considered in questions 3 and 4, which one has the highest branch prediction accuracy for the given branch outcome sequence? Justify your answer [15pts].

1.

For P1: 800ps x 1000 insts = 800ns

For P2: (7x200ps) + (200ps x 999 insts) = 201.2ns

Speed up = $\frac{800}{201.2}$ = 3.976 times faster

2.

For forwarding most recently computed X9 value:

IF	ID	EXE	MEM	WB				
	IF	ID	EXE	MEM	WB			
		IF	ID	EXE	MEM	WB		
			bubble	bubble	bubble	bubble	bubble	
				IF	ID	EXE	MEM	WB

(a) Data dependencies:

Inst 1 to Inst 2
Inst 1 to Inst 3
Inst 2 to Inst 3
Inst 2 to Inst 4
Inst 3 to Inst 4

Data hazards:

Inst 1 to Inst 2
Inst 2 to Inst 3
Inst 3 to Inst 4

(b) As shown in multi-cycle pipeline diagram above,
EXE forwarding path - For ADD instructions (1 & 2)
MEM forwarding path - For LDUR instruction (3) as 1 stall cc required to access memory.

3.

Outcome	T	T	N	N	N	T	N	T	N	T	T	N	N
Prediction	T	T	T	N	N	N	T	N	T	N	T	T	N
Hit/Miss	H	H	M	H	H	M	M	M	M	M	H	M	H

Branch prediction accuracy = 6/13

4. Static prediction strategies:

Accuracy for "always taken" = 6/13

Accuracy for "always not taken" = 7/13

⇒ "Always predict not taken" static branch prediction strategy has the highest branch prediction accuracy.